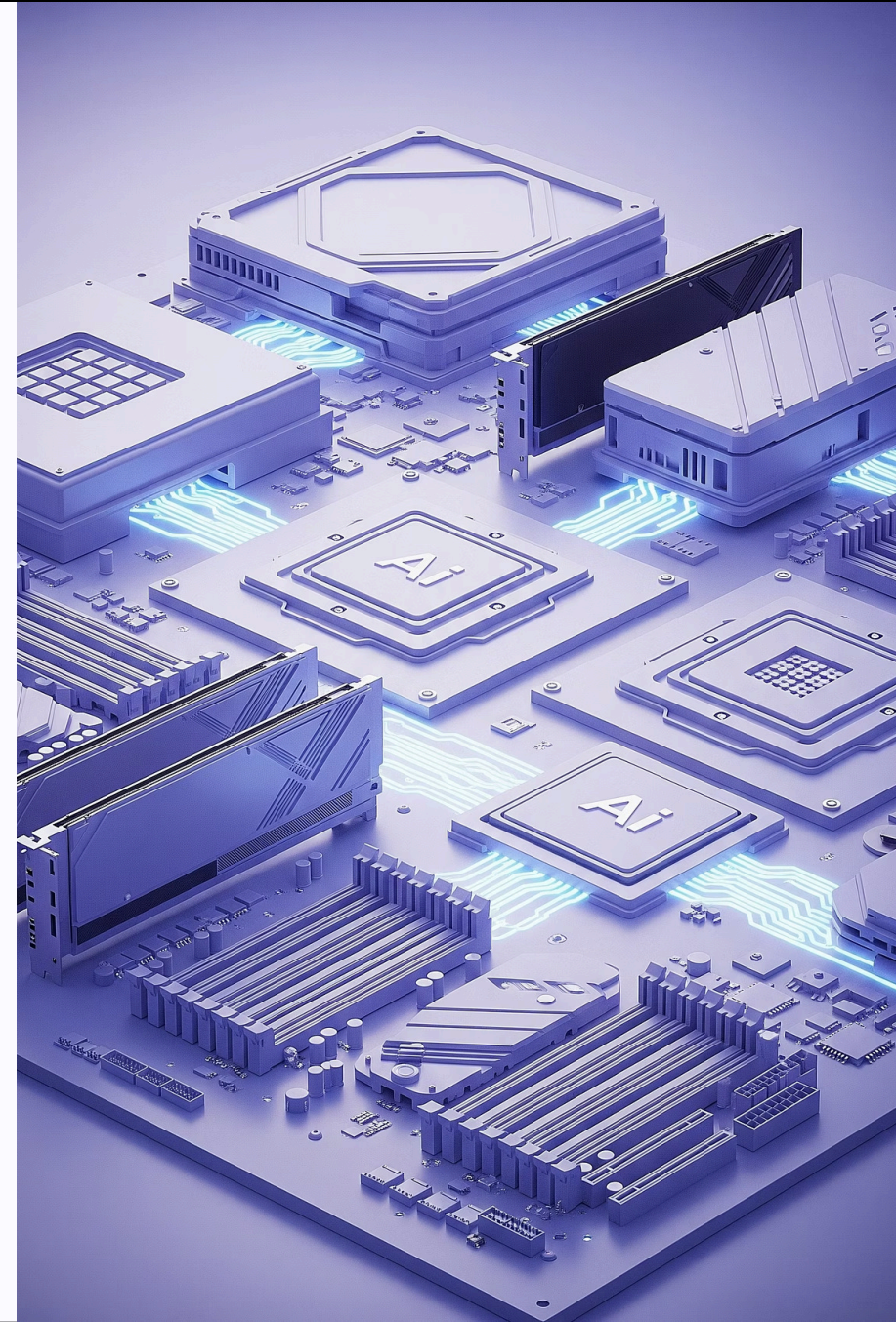


CPU vs GPU vs TPU: Computing Power for AI

A comprehensive comparison of compute architectures for machine learning workloads – helping you make informed infrastructure decisions for AI training and deployment.



Introduction to AI Compute Hardware

AI workloads demand **specialized compute resources** that go beyond traditional CPU processing capabilities. Your hardware selection directly impacts:

Performance & Training Speed

The right architecture can reduce training time from weeks to days or hours

Cost Efficiency

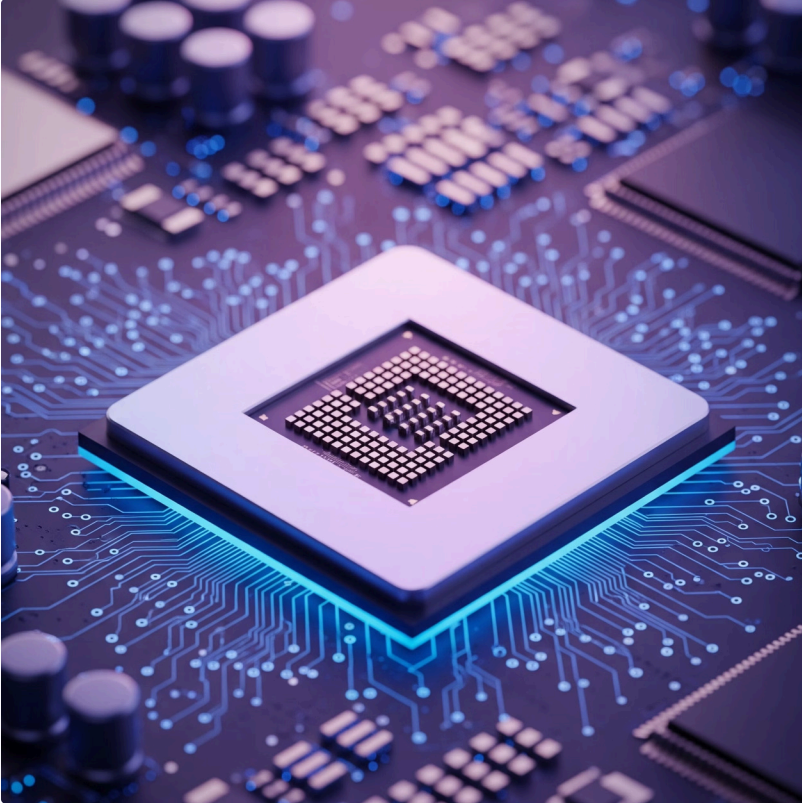
Hardware choice can mean 10-100x difference in price-performance ratio

Scalability

Different architectures scale differently with model size and data volume



CPUs – General Purpose Workhorses



Central Processing Units

Designed for **sequential processing** with high clock speeds and sophisticated control logic, CPUs excel at general-purpose computing tasks.

Ideal Use Cases:

- Data preprocessing and ETL pipelines
- ML orchestration and workflow management
- Low-latency API serving for simple models
- Small-scale inference workloads

❏ **Trade-off:** While offering unmatched versatility and software compatibility, CPUs struggle with the parallel matrix operations that dominate deep learning workloads.

GPUs – Parallel Processing Powerhouses



Parallel Architecture

Thousands of simpler cores designed for simultaneous operations on large data sets



Matrix Math Acceleration

Specialized for the tensor operations that dominate neural network training



High Memory Bandwidth

Optimized for moving massive data volumes required for AI training

Industry Standard for Deep Learning

GPUs have become the dominant hardware for training and deploying complex models across computer vision, NLP, and generative AI.

Key Examples:

- **NVIDIA H100:** Powers most frontier LLM training
- **NVIDIA A100:** Workhorse for production AI systems
- **AMD MI300X:** Emerging competitor with high memory capacity



TPUs – AI-Specific Accelerators

Tensor Processing Units

Custom-designed by Google specifically for deep learning workloads, TPUs represent the most specialized AI hardware available in cloud environments.

"TPUs are purpose-built accelerators with matrix units optimized for tensor operations in machine learning, offering 2-3x better price-performance than comparable GPUs for certain workloads."

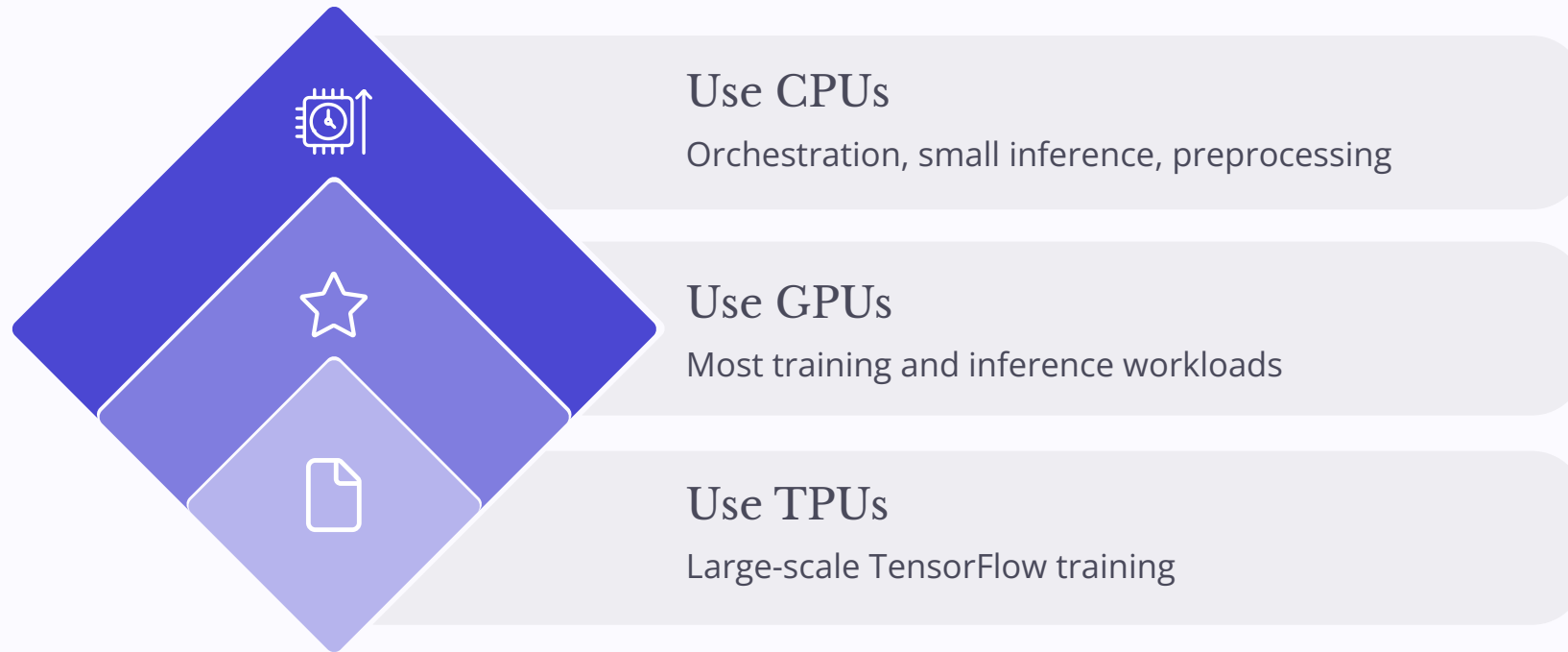
Available primarily through Google Cloud Platform, TPUs provide exceptional efficiency for TensorFlow and JAX-based workflows.



TPU Advantages:

- Custom matrix multiplication units (MXUs)
- Optimized for TensorFlow and JAX frameworks
- High-bandwidth memory (HBM)
- Scalable pod architecture for massive workloads

Choosing the Right Hardware



1

Evaluate Your Workload

Consider model size, framework, batch size, and latency requirements

2

Consider Hybrid Architectures

Most production ML systems use CPUs for preprocessing and orchestration with GPUs/TPUs for computation

3

Factor in Total Cost

Include hardware, power, cooling, engineering time, and operational complexity

The optimal solution often involves a **strategic mix** of compute types tailored to your specific AI workflow and deployment needs.